Xiaoqing PAN

xiaoqing.pan@uni-jena.de | LinkedIn | ORCID | Website | +49 15563309561 | Jena, Germany

Dear Dr. Höpfler,

I am writing to apply for the postdoctoral position in your group. My research interests focus on RNA biology and gene regulation, particularly how cellular responses are shaped through post-transcriptional mechanisms. Your work on peptide-mediated mRNA decay represents an exciting new paradigm in gene regulation grabs my interest, and I would like to bring my background in RNA biology, sequencing technologies, and computational transcriptomics to this field.

I submitted my PhD thesis in Microbiology at Friedrich Schiller University Jena (Germany) and am currently awaiting my doctoral defense. My doctoral research investigated RNA regulation during human-fungal interactions, combining molecular biology with high-throughput sequencing and bioinformatics to understand how regulatory RNA pathways respond to infectious challenges. A central aspect of my research has been integrating experimental and computational approaches. I developed and maintained reproducible workflows for mRNA-seq and sRNA-seq analysis using R and Bash on HPC environment. These pipelines included differential expression analysis, alternative splicing analysis, and transcriptome-wide characterization of RNA-mediated regulatory responses, sRNA target predictions. In parallel, I have extensive experience with molecular biology techniques, including cloning, PCR and qPCR, sequencing library preparation, Northern blotting, and microbial cell culture. This combination of wet-lab and computational expertise has allowed me to move efficiently between data generation, analysis, and biological interpretation.

I am particularly attracted to your laboratory because of its interdisciplinary approach to understanding how mRNA stability is dynamically regulated during translation. The concept that the nascent peptide itself can trigger selective degradation of the encoding transcript is both intellectually fascinating and highly relevant to broader questions in gene regulation and human disease. I am thrilled by the opportunity to contribute to the development of novel sequencing methods for identifying peptide-mediated mRNA decay substrates and to combine these approaches with computational analyses to uncover underlying molecular mechanisms.

Beyond technical expertise, I highly value collaborative and curiosity-driven research environments. Throughout my PhD, I worked closely with experimental and computational researchers across multiple institutions, coordinated transcriptomic analyses within collaborative projects, and contributed to manuscript preparation and scientific discussions. I believe that my background in RNA biology, sequencing-based technologies, and computational genomics would allow me to make meaningful contributions to the project while further developing my research at the interface of molecular biology and gene regulation. I would be very excited to discuss how my experience and scientific interests align with the ongoing work in your group.

Thank you very much for your time and consideration.

Best,

Xiaoqing

Xiaoqing PAN

xiaoqing.pan@uni-jena.de | LinkedIn | ORCID | Website | +49 15563309561 | Jena, Germany

SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized the RNAi machinery of a filamentous fungal pathogen: dicer-like protein specificity, argonaute loading, and small RNA biogenesis; performed sRNA cloning, custom sRNA-seq library preparation, target prediction, and RNA secondary structure analysis
- Validated findings using Northern blot, qPCR, and functional reporter assays; generated and characterized knockout and transgenic strains via Gibson assembly
- DOI: 10.1261/rna.079350.122 and DOI: 10.1101/2024.05.24.595671

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to three fungal pathogens, characterizing PRR pathway activation, alternative splicing, and A-to-I RNA editing
- Built pipeline; wrote custom R scripts for multi-condition cohort analysis
- DOI: 10.1021/acsinfecdis.4c00598

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge, including immunostaining of A549 epithelial cells during infection assays

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- DOI: 10.1186/s42483-023-00188-8 and DOI: 10.1016/j.jplph.2022.153905

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). *X00*-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3, OsNPF2.4*, and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Rivieccio F, Visser C, Krüger T, Weaver D, **Pan X,** Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

TECHNICAL SKILLS

- **Molecular Biology**: sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics**: RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, Snakemake, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming**: R, Python, Bash, Git, Docker, Linux
- **Language**: Chinese (native), English (fluent), German (B1)

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)